

Lie Groups

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Abstract

Potentially good books: Humphreys, Erdmann and Wildson

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1 Introduction to Lie Groups

1.1 Lie Groups

Definition 1.1.1 (Lie Groups)

A Lie group G is a group G that is also a smooth manifold such that the following are true.

- The multiplication map $\cdot : G \times G \rightarrow G$ defined by

$$(g, h) \mapsto gh$$

is a smooth map of manifolds.

- The inverse map $(-)^{-1} : G \rightarrow G$ defined by

$$g \mapsto g^{-1}$$

is a smooth map of manifolds.

Some immediate examples of Lie groups include the following:

- Any finite group is discrete and hence are zero-dimensional manifolds. So they are Lie groups.
- $(\mathbb{R}^n, +)$ and $(\mathbb{R}^n \setminus \{0\}, \times)$.
- The torus $(\mathbb{R}^n / \mathbb{Z}^n, +)$.
- $GL(V)$ for any finite dimensional real vector space V .
- $U(n), SU(n), O(n), SO(n), SL(n), PSL(n)$ for any $n \in \mathbb{N}$.

Proposition 1.1.2

Let G be a Lie group. Let H be a subgroup of G . If H is closed in G , then H inherits the structure of a Lie group from G .

Definition 1.1.3 (Lie Group Homomorphism)

Let G, H be Lie groups. A Lie group homomorphism is a map of sets

$$\phi : G \rightarrow H$$

that is a group homomorphism and a smooth map.

Definition 1.1.4 (The Identity Component)

Let G be a Lie group. Define G^0 to be the connected component of G containing 1_G .

Lemma 1.1.5

Let G be a Lie group. Then the following are true.

- $G^0 \trianglelefteq G$ is a normal subgroup.
- G^0 is a Lie group.
- G/G^0 is discrete.

Definition 1.1.6 (Lie Subgroups)

Let G be a Lie group. A Lie subgroup of G is a subgroup $H \leq G$ such that the following are true.

- H is an immersed submanifold.
- H is a Lie group in its own right.

Proposition 1.1.7

Let G be a Lie group. Let $H \leq G$ be a subgroup. If H is an embedded submanifold of G , then H is a Lie subgroup of G .

Theorem 1.1.8 (Closed Subgroup Theorem)

Let G be a Lie group. Let $H \leq G$ be a closed subgroup. Then H is an embedded submanifold of G .

Thus any closed subgroup of a Lie group is automatically a Lie subgroup.

Proposition 1.1.9

The following are true regarding the determinant map.

- $\det : GL_n(\mathbb{R}) \rightarrow \mathbb{R}$ is a Lie group homomorphism.
- The map $(\det_*)_I : T_I(GL_n(\mathbb{R})) \rightarrow T_1(\mathbb{R})$ is given by $(\det_*)_I(X) = \text{tr}(X)$.

Theorem 1.1.10 (The First Isomorphism Theorem for Lie Groups)

Let G, H be Lie groups. Let $f : G \rightarrow H$ be a Lie group homomorphism. If $\text{im}(G)$ is a closed subgroup of H , then f induces an isomorphism of Lie groups

$$\frac{G}{\ker(f)} \cong \text{im}(G)$$

1.2 Relation between Lie Groups and Lie Algebras

For a group G , denote the left multiplication map of $h \in G$ by l_h . If G is a Lie group, we have seen that l_h is a smooth map, and so it induces a differential $(l_h)_*$.

Definition 1.2.1 (Left Invariant Vector Field)

Let G be a Lie group and X a vector field on G . We say that X is left invariant if

$$(l_h)_*(X_g) = X_{hg}$$

for all $X_g \in T_g(G)$.

Proposition 1.2.2

Let G be a Lie group. The vector space of left invariant vector fields of G is a Lie algebra of dimension $\dim(G)$.

Definition 1.2.3 (Lie Algebra of a Lie Group)

Let G be a Lie group. Define the Lie algebra of G to be $\mathbb{R}$ -vector space

$$\text{Lie}(G) = \{X : G \rightarrow TG \mid X \text{ is a left invariant vector field}\}$$

together with the Lie bracket on vector fields.

Proposition 1.2.4

Let G be a Lie group. Then there is an $\mathbb{R}$ -vector space isomorphism

$$\text{Lie}(G) \cong T_{1_G}(G)$$

given by $X \mapsto X(1_G)$.

Recall that given a homomorphism of Lie groups $\phi : G \rightarrow H$, it induces a differential $\phi_* : T_g(G) \rightarrow T_{\phi(g)}(H)$.

Proposition 1.2.5

Let G, H be Lie groups. Let $f : G \rightarrow H$ be a Lie group homomorphism. Then the induced map

$$(f_*)_{1_G} : T_{1_G}(G) \rightarrow T_{1_H}(H)$$

is a Lie algebra homomorphism.

In other words, we constructed a functor from Lie groups to Lie algebras sending a Lie group to the tangent space at the identity.

Example 1.2.6

Let G be a closed subgroup of $GL_n(\mathbb{R})$. Then we have

$$\text{Lie}(G) = \{M \in M_n(\mathbb{R}) \mid e^{tX} \in G \text{ for all } t \in \mathbb{R}\}$$

Proposition 1.2.7

Let G, H be Lie groups. Let $f : G \rightarrow H$ be a Lie group homomorphism. Then the following are true.

- $\text{Lie}(\ker(f)) = \ker(df)$.
- If $f(G)$ is a closed subgroup, then $\text{Lie}(\text{im}(f)) = \text{im}(df)$.

1.3 The Exponential Map

Definition 1.3.1 (One Parameter Subgroups)

Let G be a Lie group. A one-parameter subgroup of G is a Lie group homomorphism $\gamma : \mathbb{R} \rightarrow G$.

Lemma 1.3.2

Let G be a Lie group. Let $X \in T_{1_G}(G)$. Then there exists a unique one-parameter subgroup $\gamma : \mathbb{R} \rightarrow G$ of G such that

$$X_\gamma = X$$

Definition 1.3.3 (The Exponential Map)

Let G be a Lie group. Define the exponential map $\exp : \text{Lie}(G) \rightarrow G$ by

$$\exp(X) = \gamma(1)$$

where $\gamma : \mathbb{R} \rightarrow G$ is the unique one-parameter subgroup of G whose tangent vector at 1_G is X .

Proposition 1.3.4

Let G, H be Lie groups. Let $f : G \rightarrow H$ be a Lie group homomorphism. Let $X \in T_{1_G}(G)$. Then we have

$$\exp(f_*(X)) = f(\exp(X))$$

Recall that a finite dimensional vector space V over $\mathbb{R}$ has the structure of a manifold by identifying V with $\mathbb{R}^n$. Since $\mathbb{R}^n$ is a group, V is a Lie group whose associated Lie algebra is then $\mathbb{R}^n \cong V$.

Proposition 1.3.5

Let G be a Lie group. Then the following are true.

- $\exp : \text{Lie}(G) \rightarrow G$ is a smooth map.
- $(\exp_*)_0 : \text{Lie}(G) \rightarrow T_{1_G}(G)$ is the identity map.
- There exist open subsets $0 \in U \subseteq \text{Lie}(G)$ and $1_G \in V \subseteq G$ such that $\exp|_U$ is a diffeomorphism.

Lemma 1.3.6

Let G be an abelian Lie group. Then $\exp : \text{Lie}(G) \rightarrow G$ is a Lie group homomorphism.

Proposition 1.3.7

Let G be a Lie group. Suppose that one of the following is true.

- G is connected and compact.
- G is connected and nilpotent.
- $G = GL_n(\mathbb{R})$.

Then $\exp : \text{Lie}(G) \rightarrow G$ is surjective.

1.4 The Correspondence between Lie Groups and Lie Algebras

Theorem 1.4.1 (Lie's Third Theorem)

Let $\mathfrak{g}$ be a finite dimensional Lie algebra. Then there exists a simple connected Lie group G such that $\text{Lie}(G) \cong \mathfrak{g}$.

Theorem 1.4.2 (The Homomorphisms Theorem)

Let G, H be Lie groups such that G is simple connected. Let $\phi : \text{Lie}(G) \rightarrow \text{Lie}(H)$ be a Lie algebra homomorphism. Then there exists a unique Lie group homomorphism $f : G \rightarrow H$ such that $\phi = df$.

Abstract nonsense: The functor F from the category of connected Lie groups to the category of finite dimensional Lie algebras admit a left adjoint Γ whose unit map $\mathfrak{g} \rightarrow F(\Gamma(\mathfrak{g}))$ is an isomorphism, and whose counit map $\Gamma(F(G)) \rightarrow G$ is the universal covering space for G .

Theorem 1.4.3 (The SUBgroups-Subalgebras Theorem)

Let G be a Lie group. Let $\mathfrak{h} \leq \text{Lie}(G)$ be a Lie subalgebra. Then there exists a unique connected Lie subgroup $H \leq G$ such that $\mathfrak{h} = \text{Lie}(H)$.

2 Maximal Tori

2.1 Connected Abelian Lie Groups

Proposition 2.1.1

Let G be a connected abelian Lie group. Then $\exp : \text{Lie}(G) \cong \mathbb{R}^{\dim(G)} \rightarrow G$ is the universal covering space of G .

Proposition 2.1.2

Let G be a connected abelian Lie group. Then there is a Lie group isomorphism

$$G \cong T^k \times \mathbb{R}^{\dim(G)-k}$$

for some $k \in \mathbb{N}$.

2.2 Properties of Tori

Corollary 2.2.1

Let G be a compact connected abelian Lie group. Then there is a Lie group isomorphism $G \cong T^{\dim(G)}$.

Definition 2.2.2 (Generator of a Torus)

Let $t \in T^k$. We say that t is a generator of T^k if the smallest closed subgroup containing t is T^k .

Proposition 2.2.3

Every torus admits a generator.

Proposition 2.2.4

Let $t = [(t_1, \dots, t_k)] \in (\mathbb{R}/\mathbb{Z})^k$. Then t is a generator if and only if $1, t_1, \dots, t_k$ are linearly independent over $\mathbb{Q}$.

2.3 Maximal Tori

Definition 2.3.1 (Maximal Torus)

Let G be a compact Lie group. A maximal torus of G is a Lie subgroup $T \leq G$ such that the following are true.

- T is a torus.
- T is not properly contained in a larger Lie subgroup of G that is a torus.

Proposition 2.3.2

Let G be a compact connected Lie group. Let T be a maximal torus of G . Then the following are true.

- If $T' \leq G$ is another maximal torus of G , then T and T' are conjugates in G .
- We have

$$G = \bigcup_{T \text{ is a maximal torus of } G} T$$

- We have

$$Z(G) = \bigcap_{T \text{ is a maximal torus of } G} T$$

2.4 The Weyl Group

3 Representation Theory of Lie Groups

4 Homotopy Theory of Lie Groups

Proposition 4.0.1

Let G be a connected Lie group. Then the following are true.

- G admits a universal covering space $\tilde{G}$ that is a smooth manifold.
- The structure map $p : \tilde{G} \rightarrow G$ is a morphism of Lie groups.
- We have $\pi_1(G) \cong \ker(p)$.